

Statistical Learning

Group assignment

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For this assignment you will work in **groups of 4 students**. It is ok to work in a group of 3 (for example, to simplify planning or logistics if you cannot be at university too often). However, note that a group of 3 will of course increase the individual workload because the total work is split among fewer people. It is not allowed to do this assignment individually or in a group of 2, as this goes against the spirit of working in a group.

It is up to you to find your partners. You can sign up for a group on Canvas.

Perform the tasks below and prepare a report. Do not address the questions individually point-by-point, but instead tell the whole story in a properly structured report that contains

- a brief description of your implementation (with a focus on the aspects or properties that are relevant for your simulations or for addressing the questions below),
- a discussion of your simulations (including a complete and clear description the data generating process),
- an intuitive presentation and clear discussion of your results.

Furthermore, motivate any choices that you make in your analyses. **Be concise** and use as few pages as necessary, your report may be **no longer than 8 pages** (excluding references). It is possible to add an appendix (for instance with additional results), but you will only be graded on the main text of your report (so make sure that the main text can be understood without going through the appendix). **Do not include extra “fluff” like an introduction or a separate conclusions section** where you repeat main findings.

Please submit your report in pdf format **via Canvas**, together with two R files (one containing your implementation of the method, one with the code for your simulations). Note that you are not graded on the quality of your code other than its correctness.

The deadline for submission is Friday, **October 9, 2026, 18:00**.

Assignment

In the lectures, we have discussed donor-based imputation methods and model-based imputation methods. Of course, hybrid versions can be used as well. For instance, we can incorporate model-based weights when computing distances in k -nearest-neighbor (k NN) imputation.

For simplicity, consider only continuous variables.

1. Implement a hybrid k NN imputation algorithm as follows. For imputing each variable, do the following:

- Use a weighted distance with weights based on a variable importance measure from a random forest model. Should you standardize the variables in the data?
- Recall that distances between two observations are computed using only variables that are observed in both variables. Make sure that the weights of the variables that are actually used sum up to 1. (See also the solution to Exercise 2 from Week 1 for a detailed discussion.) Why is this relevant?
- Choose a distance measure and an aggregation method that is suitable for continuous data.

On Canvas, you can find an R file containing a code skeleton and some explanations. Please use this file, **do not change the function name and make sure that you use the correct input and output as specified.**¹

Submit the R file with your implementation together with your report and your simulation code.

Your function should be written in a general manner so that it works correctly on a data set that you haven't seen (and without throwing errors).

You may use an existing package for the random forest and a corresponding variable importance measure, but you have to implement the modified k NN algorithm yourself.

2. Using your implementation, compare the standard k NN algorithm (without weights) and the hybrid k NN algorithm (with model-based weights) in a simulation study.

- Choose suitable data generating processes and make sensible choices for the number of observations n and the number of variables p . For a fair comparison, you should investigate two scenarios: one where unweighted distances are expected to perform well, and one where distances with model-based weights are expected to perform well. Do the results match your expectations?
- Investigate different missing data mechanisms: MCAR, MAR, and MNAR. You don't have to generate missing values in all variables, but there should be more than one variable with missing values. Control the proportion of missing values in each variable (and make sensible choices).

Do not use existing functions such as `ampute()` from package `mice`, but instead implement the missing data mechanisms yourself.

- Based on your choice of distance measure and aggregation method in (hybrid) k NN, which evaluation metric do you suggest to use for assessing the quality of the imputations?

Submit the R file with your simulation code together with your implementation of hybrid k NN and your report.

If necessary (to maintain the page limit), make sensible choices as to which results to report in the main text of your report and which results to put in an appendix.

¹This will simplify grading because each group's code will be similarly structured. **Not using the specified input or output format could lower your grade if your code cannot be tested properly.**

Use of AI

We handle a similar AI policy as in other courses. You may use AI to help with various tasks (e.g., implementation of the method, designing the simulation study, polishing writing). But it is expected that you carefully check and correct its output. It is not allowed hand in fully AI generated code and a fully AI written report. You are responsible for your submission. Think of AI as a fast and hard-working but unreliable team member.

Moreover, it is expected that you declare your use of AI in a statement. You can put this statement in a separate section at the end of your report. The AI statement doesn't count towards the page limit for your report. Use of AI without declaration will be considered fraud and reported to the examination board.